

UNIT 1

Introduction to Project Management

Project: AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms

1.1 Experiment 1: Project Requirement Gathering and Analysis

1.1.1 Software Project Management and Scope

Software project management encompasses the systematic planning, organizing, directing, monitoring, and controlling of software development projects to ensure they are completed on time, within budget, and to the required quality standards. It bridges the gap between technical development and business objectives by applying structured methodologies to manage people, processes, and technology effectively.

The AI-Based Bead Numeration Simulation System project scope encompasses the full-cycle development of an AI-powered educational web application that teaches users about the ancient Indian numeration system, the mathematical significance of zero, the evolution of number systems (decimal, octal, hexadecimal, binary), and the foundational role of Indian mathematics in modern digital computing. The system leverages Next.js for the frontend, Firebase for backend services, Gemini LLM for AI-driven story narration, and is deployed on Vercel. The project is managed under a structured five-phase plan spanning January 2026 to May 2026, executed by a three-member team.

1.1.2 Requirement Engineering and Elicitation

Requirement engineering is the disciplined process of eliciting, analyzing, documenting, prioritizing, and validating the needs and constraints of all stakeholders before and during software development. It ensures that the final system addresses real user needs and aligns with the project's academic and educational objectives.

For this project, requirement gathering was conducted using a hybrid approach combining facilitated workshops, document analysis, and competitive analysis. The team members included Ashray Singh (Project Manager), Madhav Nehra (Full-Stack Developer), and Rudransh Yadav (AI and Content Engineer), along with the faculty advisor for academic validation.

1.1.3 Functional Requirements

ID	Requirement Description	Source	Priority
FR01	AI-powered story narration of Indian numeration history using Gemini LLM	Workshop	High
FR02	Interactive abacus simulator with bead manipulation for addition and subtraction	Workshop	High
FR03	Ancient Indian calculation methods module (Vedic Math tricks, historical methods)	Document Analysis	High
FR04	Number system explorer with conversion tool (Decimal ↔ Octal ↔ Hex ↔ Binary)	Workshop	High
FR05	Zero and Binary connection module explaining Boolean logic and digital circuits	Document Analysis	High
FR06	Quiz and assessment module for student self-evaluation	Workshop	Medium
FR07	User authentication and role-based access using Firebase Auth	Technical Review	Medium
FR08	Admin and content management dashboard for educators	Workshop	Medium
FR09	Progress tracking for registered students across modules	Technical Review	Medium
FR10	Responsive UI accessible on desktop and mobile browsers	Document Analysis	High

1.1.4 Non-Functional Requirements

Category	Requirement	Target Metric	Priority
Performance	Page load time for all modules	< 3 seconds	High
AI Response Time	Gemini narration generation latency	< 5 seconds	High
Scalability	Concurrent users supported	Up to 500	Medium
Availability	Platform uptime	99% uptime	Medium
Security	Firebase Auth data protection	Role-based access	High
Maintainability	Modular codebase for updates	New module in 3 person-days	Medium
Usability	Learning curve for new student users	< 30 minutes	High
Compatibility	Browser and device compatibility	Chrome, Firefox, Edge	High
Reliability	Mean Time Between Failures	> 150 hours	Medium

1.1.5 Requirement Prioritization using MoSCoW Method

The MoSCoW prioritization framework was applied to classify all requirements into Must Have, Should Have, Could Have, and Won't Have categories based on delivery criticality and timeline constraints.

Must Have requirements included AI story narration, interactive abacus simulator, number system explorer, zero-binary module, user authentication, and responsive UI. Should Have requirements included Vedic Math methods, quiz module, and progress tracking. Could Have features included multilingual support, gamification, and offline mode. Mobile-native apps and ERP integrations were considered out of scope.

1.1.6 Requirements Traceability Matrix

Req ID	Source	Business Driver	Validation Method
FR01	Workshop	AI-driven educational engagement	Narration test with students
FR02	Workshop	Hands-on ancient tool simulation	20 bead operation scenarios
FR03	Document Analysis	Preserving historical computation methods	Faculty expert review
FR04	Workshop	Bridge ancient and modern number systems	50 conversion test cases
FR05	Document Analysis	Link Indian mathematics to digital architecture	Domain review
FR06	Workshop	Student self-assessment	UAT with student users
FR07	Technical Review	Secure multi-role access	Firebase Auth integration test
FR08	Workshop	Content management for educators	Admin panel walkthrough
FR09	Technical Review	Track student learning progress	Firestore test with records
FR10	Document Analysis	Broad device accessibility	Cross-browser testing

1.1.7 Project Scope Management

In scope are the AI narration module, interactive abacus simulator, Vedic mathematics content, number system explorer, zero and binary module, quiz and self-assessment, Firebase authentication, Vercel deployment, and complete user documentation. Out of scope are native mobile app development, live tutoring, LMS integration, OCR features, and curriculum beyond the defined modules.

1.2 Experiment 2: Project Identification Process Methodology and Stakeholder Description

1.2.1 Project Identification and Selection

Phase	Activity	Outcome for This Project
Problem Identification	Analysis of gaps in educational tools	Absence of AI-interactive Indian numeration tools identified
Opportunity Analysis	Assessment of AI storytelling potential	Gemini suitable for historical narration
Feasibility Study	Technical, economic, and operational assessment	Project found feasible
Project Justification	Alignment with educational objectives	Supports IT Project Management curriculum
Approval Gate	Faculty submission and approval	Approved by faculty advisor

1.2.2 Feasibility Study Details

Technical feasibility was confirmed through the compatibility of Next.js, Firebase, Gemini API, and Vercel deployment. Economic feasibility estimated the total project cost between INR 8–12 Lakhs, while operational feasibility was supported by the availability of a skilled three-member team and a faculty advisor.

1.2.3 Initiation Phases and Contracting

The project initiation involved Project Charter development, stakeholder identification, role assignment, technology stack agreement, and timeline commitment. Ashray Singh was designated as Project Manager, Madhav Nehra as Full-Stack Developer, and Rudransh Yadav as AI and Content Engineer.

1.2.4 Stakeholder Analysis

Stakeholder	Role in Project	Interest Level	Power / Influence	Expectation
Faculty Advisor	Academic oversight and grading	High	High	On-time delivery and proper documentation
Ashray Singh	Planning, scheduling, reporting	High	High	Successful project delivery
Madhav Nehra	Frontend and backend development	High	Medium	Clear requirements and architecture
Rudransh Yadav	Gemini integration and educational content	High	Medium	Well-defined AI scope
Students	Primary learners using platform	High	Low	Engaging and accurate content
Educators	Secondary users managing content	Medium	Low	Easy content management
College/Institution	Academic value and compliance	Medium	High	Demonstrated learning outcomes

1.2.5 Power-Interest Grid Classification

Quadrant	Stakeholders	Engagement Strategy
High Power, High Interest	Faculty Advisor, College	Manage Closely
High Power, Low Interest	College Administration	Keep Satisfied
Low Power, High Interest	Students, Educators	Keep Informed
Low Power, Low Interest	General Public	Monitor

1.2.6 Stakeholder Engagement Strategy

Stakeholder	Engagement Method	Frequency
Faculty Advisor	Milestone review + report submission	Bi-weekly
Ashray Singh	Internal team stand-up	Daily
Madhav Nehra	Sprint planning + code review	Per Sprint
Rudransh Yadav	Content and AI validation sessions	Weekly
Students	Beta testing + feedback surveys	Per module release
Educators	Demo walkthrough + usability feedback	At v1.0 release

1.3 Experiment 3: Project Cost Estimation and Capital Budgeting

1.3.1 Cost Components Breakdown

Cost Category	Description	Estimated Range (INR)
Development	Frontend, backend, AI integration, module development	4–6 Lakhs
AI & Content	Gemini usage and content design	1–1.5 Lakhs
Infrastructure	Vercel hosting, Firebase plan, domain	0.5–1 Lakh
Tools & Licenses	GitHub Pro, Figma, tools	0.25–0.5 Lakhs
Testing & QA	Functional and cross-browser testing	0.5–1 Lakh
Project Management	Documentation and reporting	0.5 Lakh
Contingency (15%)	Unforeseen expenses	1–1.5 Lakhs
Total		8–12 Lakhs

1.3.2 Software Cost Estimation Models

COCOMO Intermediate Organic Mode was used because this is a relatively small project with a three-member team operating in a familiar technology environment. Using a midpoint estimate of 20 KLOC, the project effort was estimated at approximately 78.6 person-months before adjustment. A practical adjusted effort range of 40–50 person-months was adopted for the academic setting with parallel task execution.

Function Point Analysis produced an adjusted value of approximately 127 FP, which was further interpreted as 150–180 FP considering AI complexity and simulation-heavy modules. Expert judgment from the team validated a budget estimate of INR 8–12 Lakhs.

1.3.3 Capital Budgeting: Net Present Value (NPV)

Parameter	Value
Discount rate	10% per annum
Project life	5 years
Initial investment	INR 10 Lakhs
Annual operating costs	INR 1.5 Lakhs
NPV	INR +28.38 Lakhs
Payback Period	~14 months
IRR	~78%

1.4 Experiment 4: Market Demand Analysis and Demand Planning

1.4.1 Market Demand Drivers

The market demand for this project is supported by the growth of the Indian EdTech market, the digital education emphasis of NEP 2020, the lack of AI-powered heritage education tools, and increasing academic interest in Indian mathematical heritage and the history of computing.

1.4.2 Target Market Segments

Segment	Demand Level	Estimated Market Size
Schools (CBSE/ICSE, Grades 6–12)	High	25,000+ schools
Engineering & Computer Science Colleges	High	4,500+ institutions
Digital Heritage Organizations	Medium	50+ organizations
EdTech Platforms	Medium	15–20 major platforms
Independent Learners	High	Millions

1.4.3 TAM Estimation by Year

Year	Target	Revenue (INR)	Market Penetration
Year 1	Free pilot to 5 institutions	0	Pilot Phase
Year 2	10 paid institutions	7.5 Lakhs	2%
Year 3	25 institutions + 3 API partners	22.75 Lakhs	5%
Year 4	50 institutions + 8 API subscribers	43.5 Lakhs	10%
Year 5	80 institutions + partnerships	72 Lakhs	15%

1.4.4 Demand Planning Strategy

Phase	Timeline	Activities & Targets
Academic Adoption	Year 1	Free pilot to schools and colleges; collect feedback
Institutional Licensing	Years 2–3	Paid licensing; 25 institutions by Year 3
Scale & Partnership	Years 4–5	EdTech partnerships; expanded content; higher revenue

1.4.5 Competitive Analysis

Competitor	Strengths	Weaknesses	Threat Level
Khan Academy	Free content, large user base	No Indian heritage focus	Low
Brilliant.org	Interactive learning	Paid, no ancient Indian math content	Low
Vedic Maths Forum	Deep Vedic content	No AI, outdated UI	Low
IGNCA Digital Portal	Authentic cultural content	No interactive educational tools	Low
Custom School ERP Modules	Embedded in institutions	Generic content	Medium

1.4.6 Market Differentiation Strategy

The project differentiates itself through its AI-first approach, interactive simulation of ancient mathematical tools, complete historical-to-digital narrative arc, curriculum alignment, and blended free-plus-institutional licensing model.

1.4.7 Conclusion for Unit 1

Unit 1 established the foundational planning framework for the project. The requirement engineering phase defined 10 functional and 9 non-functional requirements. Stakeholder and feasibility analysis confirmed that the project is technically, economically, and operationally viable. Cost estimation through COCOMO, Function Point Analysis, and expert judgment placed the project budget in the INR 8–12 Lakhs range. Capital budgeting indicated a positive NPV and short payback period, while market analysis demonstrated strong educational and commercial potential for the platform.